

TASK-2

SENSOR BASED SIMULATION

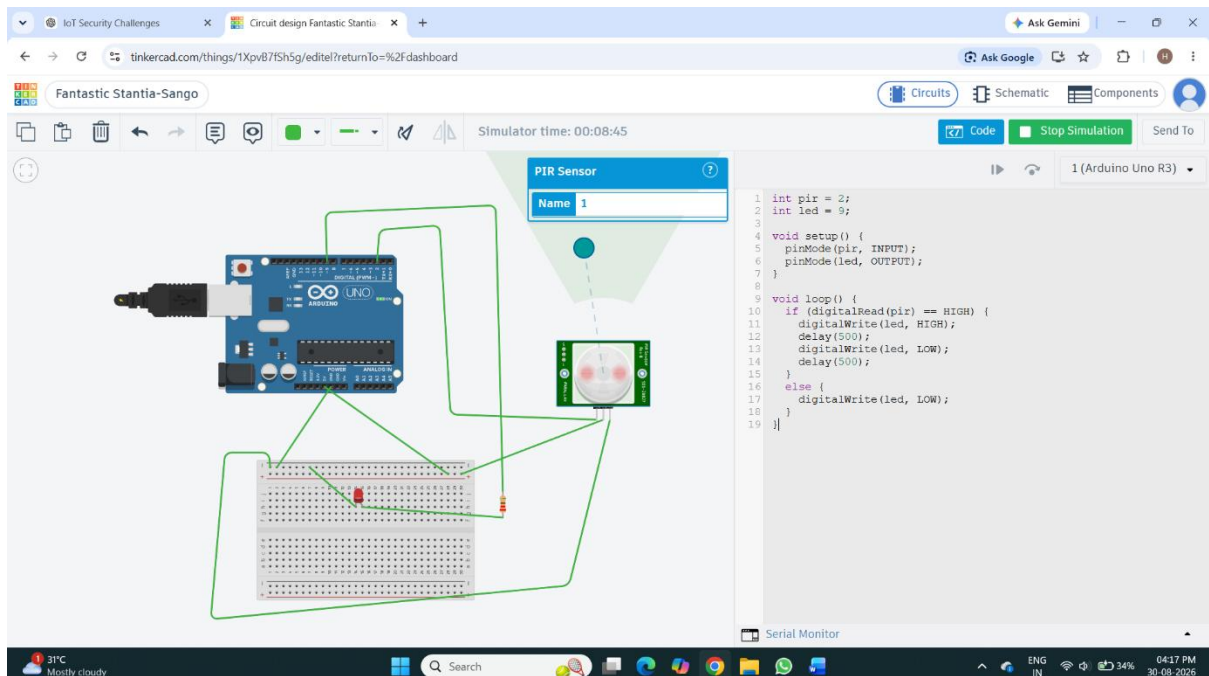
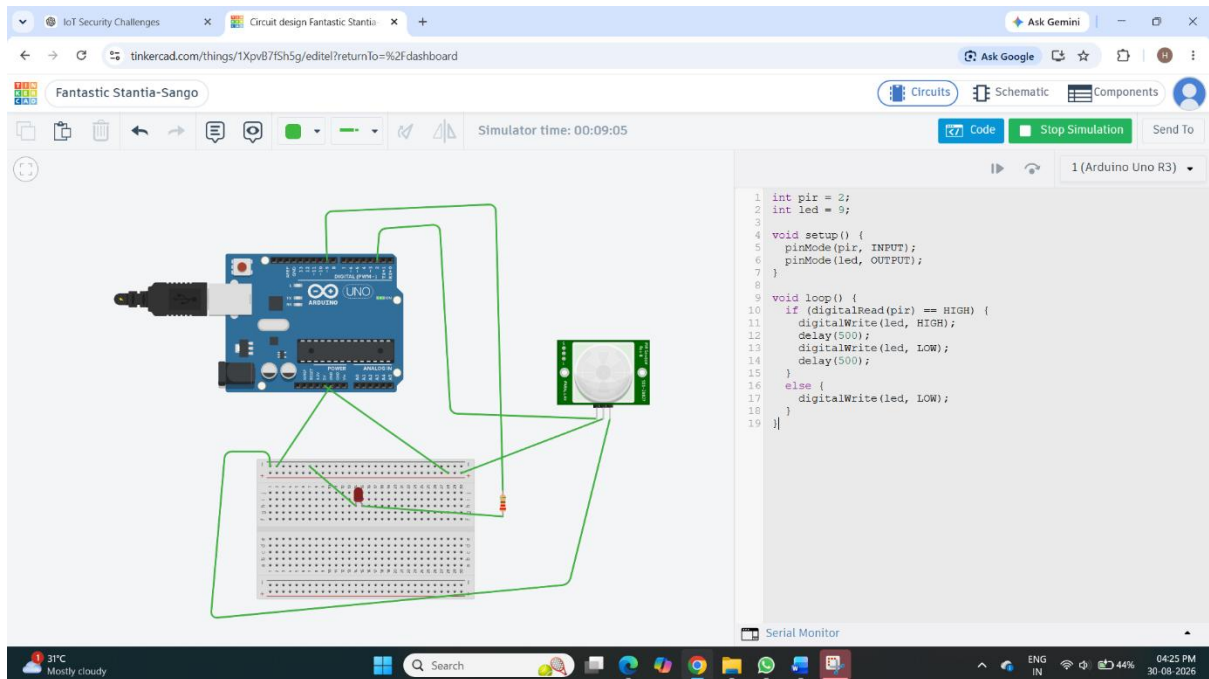


Fig: LED Blinking

1.CODE

```
int pir = 2;
```

```
int led = 9;
```

```
void setup() {
```

```
    pinMode(pir, INPUT);
```

```
    pinMode(led, OUTPUT);
```

```
}
```

```
void loop() {
```

```
    if (digitalRead(pir) == HIGH) {
```

```
        digitalWrite(led, HIGH);
```

```
        delay(500);
```

```
        digitalWrite(led, LOW);
```

```
        delay(500);
```

```
    }
```

```
    else {
```

```
        digitalWrite(led, LOW);
```

```
    }
```

```
}
```

2.CODE EXPLANATION – PIR SENSOR AND BLINKING LED

1.Pin Declaration

```
int pir = 2;
```

```
int led = 9;
```

- PIR sensor is connected to D2.

- LED is connected to D9.

2. Setup Function

```
void setup() {
  pinMode(pir, INPUT);
  pinMode(led, OUTPUT);
}
```

- PIR is set as an INPUT because it sends a signal to Arduino.
- LED is set as an OUTPUT because Arduino controls the LED.

3. Checking Motion

```
if (digitalRead(pir) == HIGH)
```

Arduino checks the PIR sensor.

- **HIGH** → Motion detected.
- **LOW** → No motion.

4. Blinking the LED

```
digitalWrite(led, HIGH);
delay(500);
digitalWrite(led, LOW);
delay(500);
```

When motion is detected:

- LED turns ON.
- Waits for 500 milliseconds.
- LED turns OFF.
- Waits for 500 milliseconds.
- This repeats, so the LED blinks.

5. No Motion

```
else {
```

```
digitalWrite(led, LOW);  
}
```

When there is no motion, the LED stays OFF.

3.COMPONENTS USED

1. Arduino Uno – Controls the circuit and processes the PIR sensor signal.
2. PIR Motion Sensor – Detects movement of people or objects.
3. LED – Indicates motion by blinking.
4. 220 Ω Resistor – Protects the LED from excessive current.
5. Breadboard – Used to connect the components without soldering.
6. Jumper Wires – Used to make electrical connections between the components.

4.WORKING PRINCIPLE

1. Power supply
 - The Arduino Uno supplies 5V to the PIR sensor.
 - The PIR sensor gets power through VCC and GND.
2. PIR detects motion
 - The PIR sensor detects movement of a person or object by sensing changes in infrared (IR) radiation.
 - When no motion is detected, the PIR OUT pin remains LOW.
 - When motion is detected, the OUT pin becomes HIGH.
3. Arduino receives the signal
 - The PIR OUT pin is connected to D2 of the Arduino.

- The Arduino continuously checks the signal on D2 using: `digitalRead(pir)`

4. Arduino controls the LED

- If D2 becomes HIGH, the Arduino knows that motion has been detected.
- It then turns the LED connected to D9 ON.

5. LED blinks

- The program keeps the LED ON for 500 ms and OFF for 500 ms.
- Therefore, the LED continuously blinks while motion is detected.

6. When motion stops

- The PIR output eventually becomes LOW.
- The Arduino then turns the LED OFF.